Math 110 — Week 2

Linear independence, span, bases, dimension (Notes Ch. 3–4, LADR 2A–2C, HW2)

§1 Facts used from Week 1

(W1) A subset $U \subseteq V$ is a subspace if $0 \in U$ and $\lambda v + w \in U$ for all $v, w \in U$, $\lambda \in \mathbb{F}$. Intersections and sums $U + W = \{u + w\}$ of subspaces are subspaces.

(W2) The sum $U_1 + \cdots + U_m$ is direct if $U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\}$ for each k. Equivalent: the only way to write $0 = u_1 + \cdots + u_m$ with $u_k \in U_k$ is with all $u_k = 0$ (Prop. 7). For two subspaces: $U \cap W = \{0\}$.

(W3) If $av = 0$ then $a = 0$ or $v = 0$.

§2 This week

§2.1 Linear independence, span, basis (Ch. 3)

The question. When does a finite list of vectors describe a whole vector space: every vector can be built from the list, and in only one way? Which of these lists in $\mathbb{R}^2$ do that: $(1,0), (0,1)$; $(1,0), (0,1), (1,1)$; $(1,0)$?

Definition (Def. 7, 8 / LADR 2.4, 2.15, 2.26, 2.29)**.** Let $v_1, \ldots, v_m \in V$. A **linear combination** is a sum $\lambda_1 v_1 + \cdots + \lambda_m v_m$ with $\lambda_k \in \mathbb{F}$.

- $\operatorname{span}(v_1, \ldots, v_m)$ is the set of all linear combinations.
- The list **spans** V if $\operatorname{span}(v_1, \ldots, v_m) = V$.
- The list is **linearly independent** if $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ forces every $\lambda_k = 0$. Otherwise it is **linearly dependent**.
- The list is a **basis** of V if it is linearly independent and spans V.
- V is **finite-dimensional** if some finite list spans it, and **infinite-dimensional** if not.
- The **length** of a list is its number of entries. The empty list has length 0, is linearly independent, and spans $\{0\}$.
- A list containing 0 is dependent: $1 \cdot 0 = 0$ with coefficient $1 \neq 0$.
- $\operatorname{span}(v_1, \ldots, v_m) = \mathbb{F}v_1 + \cdots + \mathbb{F}v_m$, where $\mathbb{F}v = \{\lambda v : \lambda \in \mathbb{F}\}$, so it is a subspace (W1). It is the smallest subspace containing $v_1, \ldots, v_m$: a subspace containing them contains each $\lambda_k v_k$ (closure under scalar multiplication) and then each sum $\lambda_1 v_1 + \cdots + \lambda_m v_m$ (closure under addition).
- **Polynomials.** For $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, $\mathcal{P}(\mathbb{F})$ is the set of functions $p\colon \mathbb{F} \to \mathbb{F}$ of the form $p(x) = a_0 + a_1 x + \cdots + a_m x^m$ with $a_k \in \mathbb{F}$; it is a subspace of $\mathbb{F}^{\mathbb{F}}$ (Week 1, Ex. 6). If $a_m \neq 0$, p has **degree** m; the zero polynomial has degree $-\infty$. $\mathcal{P}_m(\mathbb{F}) = \operatorname{span}(1, x, \ldots, x^m)$ is the subspace of polynomials of degree at most m. (Over $\mathbb{F}_2$, polynomials as functions behave differently: x^2 and x are the same function on $\{0, 1\}$.)

Example 10. $e_1, \ldots, e_n$ is a basis of $\mathbb{F}^n$. The span of $(1, 1, 1, \ldots)$ and $(0, 1, 2, 3, \ldots)$ in $\mathbb{R}^\infty$ is the set of arithmetic sequences $(a_0, a_0 + d, a_0 + 2d, \ldots)$. The only basis of $\{0\}$ is the empty list.

Proposition (Basis = unique representation, LADR 2.28)

$v_1, \ldots, v_m$ is a basis of $V \iff$ every $v \in V$ can be written as $\lambda_1 v_1 + \cdots + \lambda_m v_m$ in exactly one way.

The idea

Spanning is, by definition, the existence of a representation. Uniqueness comes from subtracting two representations of the same vector: the difference is a combination equal to 0, which is exactly what independence controls. Example: in $\mathbb{R}^2$ with the basis $(1,1), (1,-1)$, the vector $(3,1)$ is $2(1,1)+1(1,-1)$, and no other pair of coefficients works.

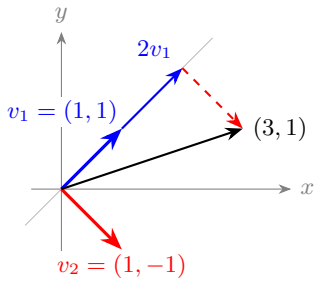
Proof. **($\Rightarrow$) Assume $v_1, \ldots, v_m$ is a basis.** Let $v \in V$.

1. *v can be written as a combination.* The list spans V, which means exactly that every vector of V is a linear combination of $v_1, \ldots, v_m$.
2. *Only one way.* Suppose $v = \lambda_1 v_1 + \cdots + \lambda_m v_m$ and also $v = \mu_1 v_1 + \cdots + \mu_m v_m$. Subtracting the two expressions,
$$0 = (\lambda_1 - \mu_1) v_1 + \cdots + (\lambda_m - \mu_m) v_m.$$
3. The list is independent, so every coefficient $\lambda_k - \mu_k$ is 0. So $\lambda_k = \mu_k$ for all k: the two ways are the same.

($\Leftarrow$) Assume every $v \in V$ can be written as a combination in exactly one way.

1. *The list spans V.* Every v can be written as a combination; that is the definition of spanning.
2. *The list is independent.* Suppose $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$. We also have $0v_1 + \cdots + 0v_m = 0$. These are two ways of writing the vector 0. By assumption there is only one way, so $\lambda_k = 0$ for all k.

□



Go 2 steps along v_1, then 1 step along v_2. Spanning says every point can be reached this way; independence says there is only one way.

- The unique coefficients $\lambda_1, \ldots, \lambda_m$ are called the **coordinates** of v in the basis.
- The notes say the same thing with the map $\Phi\colon \mathbb{F}^m \to V$, $\Phi(\lambda_1, \ldots, \lambda_m) = \lambda_1 v_1 + \cdots + \lambda_m v_m$. The list spans V exactly when Φ is surjective (every v is hit). It is independent exactly when Φ is injective: $\Phi(\lambda_1, \ldots, \lambda_m) = \Phi(\mu_1, \ldots, \mu_m)$ exactly when $(\lambda_1 - \mu_1)v_1 + \cdots + (\lambda_m - \mu_m)v_m = 0$, which for an independent list forces $\lambda_k = \mu_k$; conversely, if Φ is injective, a relation $\Phi(\lambda) = 0 = \Phi(0, \ldots, 0)$ forces $\lambda = 0$. It is a basis exactly when Φ is bijective, and then Φ^{-1} sends each vector to its list of coordinates.
- A list is independent exactly when it is a basis of its span.
- Connection with direct sums: if all $v_k \neq 0$, the list $v_1, \ldots, v_m$ is independent exactly when the sum of the lines $\mathbb{F}v_1 + \cdots + \mathbb{F}v_m$ is direct. Reason: by (W2) the sum is direct when $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ forces every *term* $\lambda_k v_k$ to be 0, and by (W3) $\lambda_k v_k = 0$ with $v_k \neq 0$ means $\lambda_k = 0$.

The question answered. A list describes V exactly when it is a basis: spanning says every vector can be built, independence says in only one way. $(1,0), (0,1)$ is a basis of $\mathbb{R}^2$; $(1,0), (0,1), (1,1)$ spans but is dependent, since $1 \cdot (1,0) + 1 \cdot (0,1) - 1 \cdot (1,1) = 0$; $(1,0)$ alone is independent but does not span.

Proposition (Prop. 8 / LADR 2.19: linear dependence lemma)

Let $v_1, \ldots, v_m$ be linearly dependent. Then

(i) there is an index j with $v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$;

(ii) for any such j, removing v_j does not change the span.

The idea

Dependence gives an equation $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ with some nonzero coefficient. Any vector with a nonzero coefficient can be solved for in terms of the others; that vector is redundant. Choosing the *last* nonzero coefficient makes all the other vectors in the equation come *earlier* in the list. That is stronger than "v_j is a combination of the others", and later proofs need it: when a list starts with vectors we want to keep, the redundant vector can be found after them. Example: for $(1,0), (0,1), (1,1), (2,0)$, the relation $1 \cdot (1,0) + 1 \cdot (0,1) - 1 \cdot (1,1) + 0 \cdot (2,0) = 0$ has its last nonzero coefficient on $(1,1)$, which is in the span of the vectors before it.

Proof. 1. Take $\lambda_1 v_1 + \cdots + \lambda_m v_m = 0$ with not all $\lambda_k = 0$. Let j be the largest index with $\lambda_j \neq 0$. The relation is $\lambda_1 v_1 + \cdots + \lambda_j v_j = 0$.

2. Divide by λ_j:

$$v_j = -\frac{\lambda_1}{\lambda_j} v_1 - \cdots - \frac{\lambda_{j-1}}{\lambda_j} v_{j-1} \in \operatorname{span}(v_1, \ldots, v_{j-1}).$$

   If $j = 1$ this says $v_1 = 0$, and the span of the empty list is $\{0\}$.

3. For (ii): write $v_j = c_1 v_1 + \cdots + c_{j-1} v_{j-1}$. In any combination $\lambda_1 v_1 + \cdots + \lambda_m v_m$, replace v_j by this expression. The result uses only the other vectors. So every vector in $\operatorname{span}(v_1, \ldots, v_m)$ is also in the span of the list without v_j. Conversely, a combination of the shorter list is a combination of the full list (give v_j the coefficient 0). So the two spans are equal.

□

Proposition (Prop. 9 / LADR 2.30: a spanning list contains a basis)

If $f_1, \ldots, f_n$ spans V, some of the f_k can be deleted so that what remains is a basis of V.

Proof. 1. If the list is independent, it is a basis.

2. If not, Prop. 8 gives a vector whose removal does not change the span. Remove it. The list still spans V and is one shorter.

3. Repeat. The list is finite, so this stops, and it stops only when the list is independent.

□

Every finite-dimensional space has a basis (LADR 2.31). By definition some finite list spans V; Prop. 9 deletes vectors from it until a basis remains.

Theorem (Prop. 10 / LADR 2.22: independent lists are not longer than spanning lists)
If $e_1, \dots, e_m$ is linearly independent in V and $f_1, \dots, f_n$ spans V, then $m \le n$.

The idea

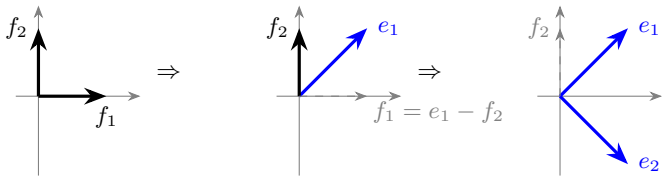
Why can $\mathbb{R}^2$ not contain three independent vectors? No single computation answers this for all lists at once, so we compare an independent list with a spanning list directly. **Exchange one vector at a time.** Start with the spanning list $f_1, \dots, f_n$ and put e_1 into it. Since the f's span, $e_1 = \lambda_1 f_1 + \cdots + \lambda_n f_n$, with some $\lambda_k \neq 0$ (because $e_1 \neq 0$), and that f_k can be solved for. So e_1 and the other $n-1$ of the f's still span. Repeat with e_2, e_3, …: each new e_k is a combination of the current list, and that combination must use some remaining f with a nonzero coefficient, since otherwise $e_1, \dots, e_k$ would be dependent. Each step replaces one f by one e. If there were more e's than f's, then after n steps no f would be left: $e_1, \dots, e_n$ would span V, so e_{n+1} would be a combination of them, contradicting independence.

Example: exchanging e_1, e_2 into f_1, f_2 in $\mathbb{R}^2$. Take $e_1 = (1,1)$, $e_2 = (1,-1)$ and $f_1 = (1,0)$, $f_2 = (0,1)$. $e_1 = f_1 + f_2$, so exchange f_1: the list e_1, f_2 spans. Then $e_2 = e_1 - 2f_2$ uses f_2, so exchange it: e_1, e_2 spans. A third independent vector is impossible.

Proof. If $m = 0$ there is nothing to prove. Claim $P(k)$, for $1 \le k \le \min(m, n)$: after relabelling the f's, the list $L_k = (e_1, \dots, e_k, f_{k+1}, \dots, f_n)$ spans V.

1. $P(1)$. Since $m \ge 1$, e_1 exists, and $e_1 \neq 0$ because an independent list does not contain 0. So $V \neq \{0\}$, and the spanning list is not empty: $n \ge 1$. Write $e_1 = \lambda_1 f_1 + \cdots + \lambda_n f_n$. Some λ is nonzero, since $e_1 \neq 0$; relabel so that $\lambda_1 \neq 0$. Solving for f_1:
$$f_1 = \frac{1}{\lambda_1}\big(e_1 - \lambda_2 f_2 - \cdots - \lambda_n f_n\big) \in \operatorname{span}(e_1, f_2, \dots, f_n).$$
So this span contains $f_1, f_2, \dots, f_n$. It is a subspace, so it contains every combination of the f's, and those are all of V.
2. $P(k-1) \Rightarrow P(k)$. Since L_{k-1} spans V, there are scalars with
$$e_k = \lambda_1 e_1 + \cdots + \lambda_{k-1} e_{k-1} + \mu_k f_k + \cdots + \mu_n f_n.$$
3. Some μ is nonzero: otherwise $e_k \in \operatorname{span}(e_1, \dots, e_{k-1})$, contradicting independence of the e's. Relabel so that $\mu_k \neq 0$.
4. Solving for f_k:
$$f_k = \frac{1}{\mu_k}\big(e_k - \lambda_1 e_1 - \cdots - \lambda_{k-1} e_{k-1} - \mu_{k+1} f_{k+1} - \cdots - \mu_n f_n\big),$$
so $f_k \in \operatorname{span}(L_k)$. Every other vector of L_{k-1} is in L_k. So $\operatorname{span}(L_k)$ is a subspace containing the spanning list L_{k-1}, hence equal to V.
5. Suppose $m > n$. Then $P(n)$ says $e_1, \dots, e_n$ spans V, so $e_{n+1} \in \operatorname{span}(e_1, \dots, e_n)$, contradicting independence. Hence $m \le n$.

The proof also shows: after all the e's are inserted, $e_1, \dots, e_m, f_{m+1}, \dots, f_n$ still spans V. □



start: f_1, f_2 | e_1 in, f_1 out | e_2 in, f_2 out

Each step puts one e in and takes one f out, and the list still spans $\mathbb{R}^2$: the removed f (dashed) is a combination of the vectors kept, $f_1 = e_1 - f_2$ and then $f_2 = \frac{1}{2}(e_1 - e_2)$.

Proposition (Prop. 11 / LADR 2.32: an independent list extends to a basis)
If V is finite-dimensional and $e_1, \dots, e_m$ is linearly independent, there is a basis of V containing $e_1, \dots, e_m$. If $f_1, \dots, f_n$ spans V, the added vectors can be taken from the f_j.

The idea

Put the two lists together: $e_1, \dots, e_m, f_1, \dots, f_n$ spans V. Now remove redundant vectors as in Prop. 9. The only worry is that an e gets removed. The dependence lemma removes a vector that is a combination of the vectors *before* it, and the e's come first and are independent, so no e is a combination of the vectors before it. Only f's are removed.

Proof. 1. The list $e_1, \dots, e_m, f_1, \dots, f_n$ spans V.

2. If it is dependent, Prop. 8 gives a vector in the span of the earlier ones. It is not some e_i, because $e_i \notin \operatorname{span}(e_1, \dots, e_{i-1})$ by independence. So it is an f_j; remove it. The span is unchanged.

3. Repeat until the list is independent. It still spans V and still contains every e_i.

□

Example 11. No three vectors in $\mathbb{F}^2$ are independent, since e_1, e_2 spans (Prop. 10). This is why three lines in $\mathbb{R}^2$ never form a direct sum.

Proposition (LADR 2A: the powers of x are independent)
Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. If $a_0 + a_1 x + \cdots + a_m x^m = 0$ for every x, then every $a_k = 0$. So $1, x, \dots, x^m$ is a basis of $\mathcal{P}_m(\mathbb{F})$, and $\mathcal{P}(\mathbb{F})$ is infinite-dimensional.

The idea

A polynomial that is 0 at every point has every derivative equal to 0 too. The k-th derivative at 0 isolates one coefficient: each term $a_j x^j$ with $j < k$ is differentiated to 0, and each term with $j > k$ still contains a factor x, which is 0 at $x = 0$.

Proof. 1. Let $p(x) = a_0 + a_1 x + \cdots + a_m x^m$ be 0 for every real x (for $\mathbb{F} = \mathbb{C}$, use only real x: the coefficients may be complex, and differentiation in the real variable x works the same way). Then every derivative $p^{(k)}$ is the zero function as well.

2. The k-th derivative of x^j is 0 if $j < k$, the constant $k!$ if $j = k$, and $j(j-1)\cdots(j-k+1)x^{j-k}$ if $j > k$, which is 0 at $x = 0$. So $p^{(k)}(0) = k!\, a_k$.

3. Since $p^{(k)}$ is the zero function, $k!\, a_k = 0$, and $k! \neq 0$ in $\mathbb{R}$ and $\mathbb{C}$. So $a_k = 0$ for every $k = 0, \dots, m$.

4. $\mathcal{P}(\mathbb{F})$ **is infinite-dimensional.** For every n, $1, x, \dots, x^n$ is an independent list of length $n+1$ in $\mathcal{P}(\mathbb{F})$ (steps 1–3). If a list of length n spanned $\mathcal{P}(\mathbb{F})$, this would contradict Prop. 10.

□

Infinite-dimensional spaces (Def. 9, optional in the notes). For an infinite set $M \subseteq V$: M is linearly independent if every finite subset is; M spans if every vector is a *finite* combination of elements of M; a basis is both.

Examples 12 and 13. **(12)** $\{1, x, x^2, \dots\}$ is a basis of $\mathcal{P}(\mathbb{R})$: every polynomial is a finite combination of powers of x, and a finite combination $a_0 + a_1 x + \cdots + a_m x^m$ is the zero polynomial only if every $a_k = 0$ (the powers of x are independent, LADR 2A).

(13) In $\mathbb{F}^\infty$, $\{e_1, e_2, \dots\}$ is independent but not a basis: $(1, 1, 1, \dots)$ is not a finite combination (HW2 P2), as the table below shows.

coordinate	1	2	3	4	5	$\cdots$
e_1	1	0	0	0	0	$\cdots$
e_3	0	0	1	0	0	$\cdots$
$2e_1 - e_3$	2	0	-1	0	0	$\cdots$
$(1, 1, 1, \ldots)$	1	1	1	1	1	$\cdots$

The combination $2e_1 - e_3$ is zero after coordinate 3. Every finite combination of the e_j is zero after its largest index j, so none equals the all-ones sequence.

The finite results have infinite-dimensional analogues, all proved with Zorn's lemma (cited; not part of the course): every independent set extends to a basis, every spanning set contains a basis (so every vector space has a basis), and there is an injective map from any independent set into any spanning set. Nobody can write down an explicit basis of $\mathbb{F}^\infty$, which is why the general proof needs Zorn's lemma.

Shortcuts

- **Last-vector test**: $v_1, \ldots, v_m$ is independent exactly when $v_1 \neq 0$ and no v_j lies in $\operatorname{span}(v_1, \ldots, v_{j-1})$. Reason: a dependent list has such a j (Prop. 8(i)); conversely $v_j = c_1 v_1 + \cdots + c_{j-1} v_{j-1}$ is a relation with coefficient -1 on v_j. Instance: $(1,0,0), (1,1,0), (1,1,1)$: the second is not a multiple of the first, and the third has third coordinate 1, while both earlier vectors have third coordinate 0.
- **Staircase lists are independent**: if each vector has a nonzero entry in a coordinate where all earlier vectors have 0, the list is independent (last-vector test). Instance: the three vectors above. For polynomials: *nonzero* polynomials of strictly increasing degrees are independent; instance $x - 1$, $x^2 - 2$ (R3).
- **Two vectors** are dependent exactly when one is a multiple of the other. Reason: in a relation $\lambda v + \mu w = 0$, $\mu \neq 0$ gives $w = -(\lambda/\mu)v$ and $\lambda \neq 0$ gives $v = -(\mu/\lambda)w$. Instance: $(2, 4) = 2(1, 2)$. For three or more vectors this test is not enough (Traps).
- **Adding a vector outside the span keeps independence**: if $v_1, \ldots, v_m$ is independent and $v \notin \operatorname{span}(v_1, \ldots, v_m)$, then $v_1, \ldots, v_m, v$ is independent. Reason: last-vector test. Instance: $(1,0,0), (0,1,0)$ and then $(0,0,1)$.
- **Sublists and superlists**: a sublist of an independent list is independent (a relation among the sublist is a relation in the full list with the other coefficients 0), and a list that contains a spanning list spans. Instance: e_1, e_3 is independent as part of e_1, e_2, e_3.
- **Equal spans**: $\operatorname{span}(v_1, \ldots, v_m) = \operatorname{span}(w_1, \ldots, w_n)$ as soon as every w_j is a combination of the v's and every v_i a combination of the w's. Reason: a span is the smallest subspace containing its vectors, so each span contains the other. Instance: $\operatorname{span}\big((1,0),(0,1)\big) = \operatorname{span}\big((1,1),(1,-1)\big)$, since $(1,0) = \frac{1}{2}(1,1) + \frac{1}{2}(1,-1)$ and $(0,1) = \frac{1}{2}(1,1) - \frac{1}{2}(1,-1)$.

Traps

- Independence of each list says nothing about sums: v and $-v$ are each independent lists (for $v \neq 0$), but $v + (-v) = 0$ (HW2 P1).
- "No vector is a multiple of another" does not give independence for three or more vectors: $(1,0), (0,1), (1,1)$, with $(1,1) = (1,0) + (0,1)$.
- A list containing 0 is never independent: $(0,0), (1,0)$ has the relation $1 \cdot (0,0) + 0 \cdot (1,0) = 0$.
- Spanning uses finite combinations only: $\{e_1, e_2, \ldots\}$ is independent in $\mathbb{F}^\infty$, but $(1, 1, 1, \ldots)$ is not a finite combination of the e_j, so the set does not span.
- Over $\mathbb{F}_2$, polynomials as functions do not have independent powers: $x^2 - x$ is the zero function on $\{0, 1\}$.

§2.2 Dimension (Ch. 4)

The question. In $\mathbb{R}^3$ let $U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$ and $W = \operatorname{span}\big((1,2,2),(2,2,1)\big)$. Without solving for their common vectors, can you tell that these two planes meet in a line, and not only in 0?

Proposition (Prop. 12 / LADR 2.34, Def. 10)
Any two bases of a finite-dimensional V have the same length. This length is the **dimension** $\dim V$.

Proof. 1. Let B_1 and B_2 be bases of V (they exist by LADR 2.31 above). Each is independent and spans V.

2. Prop. 10 with B_1 as the independent list and B_2 as the spanning list gives length $B_1 \le$ length B_2.

3. Prop. 10 with B_2 as the independent list and B_1 as the spanning list gives length $B_2 \le$ length B_1. So the lengths are equal.

□

For infinite-dimensional spaces any two bases also have the same size, but the proof needs more set theory.

Examples 14–17. **(14–16)** $\dim \mathbb{F}^n = n$, since $(x_1, \dots, x_n) = x_1 e_1 + \cdots + x_n e_n$ in exactly one way. $\dim \mathcal{P}_r(\mathbb{F}) = r + 1$ for $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, with basis $1, x, \dots, x^r$ (powers of x are independent). $\dim\{0\} = 0$.

(17) The dimension depends on the field. $\dim_{\mathbb{C}} \mathbb{C}^2 = 2$, but $\dim_{\mathbb{R}} \mathbb{C}^2 = 4$, with real basis $(1,0), (i,0), (0,1), (0,i)$. Check: $(a+bi, c+di) = a(1,0) + b(i,0) + c(0,1) + d(0,i)$, and the real numbers a, b, c, d are determined by the vector.

Proposition (Prop. 13 / LADR 2.38, 2.42: the right number of vectors)
Let $d = \dim V$. (i) Any linearly independent list of d vectors is a basis. (ii) Any spanning list of d vectors is a basis.

The idea

A basis has to pass two tests, independence and spanning. When the list already has exactly $\dim V$ vectors, passing one test is enough. We know two procedures: an independent list can be made into a basis by adding vectors (Prop. 11), and a spanning list can be made into a basis by deleting vectors (Prop. 9). The result of either procedure is a basis, so it has exactly d vectors. If the list already had d vectors, nothing was added or deleted. It is the same counting principle as the pigeonhole principle of Week 1.

Proof. 1. **(i)** Let $v_1, \dots, v_d$ be independent. By Prop. 11 we can add vectors to it (possibly none) to get a basis of V.

2. That basis has exactly d vectors, because every basis of V does (Prop. 12). The list $v_1, \dots, v_d$ already has d vectors, so no vectors were added. So $v_1, \dots, v_d$ is itself the basis.

3. **(ii)** Let $v_1, \dots, v_d$ span V. By Prop. 9 we can delete vectors from it (possibly none) to get a basis of V.

4. That basis has exactly d vectors, so no vectors were deleted. So $v_1, \dots, v_d$ is itself the basis.

□

For finite sets, $A \subseteq B$ and $|A| = |B|$ force $A = B$. The next proposition is the same statement with dimension in place of the number of elements.

Proposition (Prop. 14 / LADR 2.25, 2.37, 2.39: subspaces)
If U is a subspace of finite-dimensional V, then U is finite-dimensional, $\dim U \le \dim V$, and $\dim U = \dim V \iff U = V$.

The idea

We cannot start from a basis of U: that U has one is part of what is to be proved. What we do know is a bound: an independent list in U is independent in V, so it has at most $\dim V$ vectors (Prop. 10). So there is a longest independent list in U. A longest one must span U, because a vector of U outside its span could be added to it and the longer list would still be independent.

Proof. 1. *Independent lists in U cannot be long.* A list of vectors of U that is independent is also an independent list in V, so its length is at most $\dim V$ (Prop. 10). So among all independent lists in U there is one of largest length. Call it $u_1, \dots, u_k$; then $k \le \dim V$.

2. *This list spans U.* Suppose not: some $u \in U$ is not in $\operatorname{span}(u_1, \dots, u_k)$. We show that then $u_1, \dots, u_k, u$ is independent, which contradicts the choice of k as the largest length.
3. Take a relation $\lambda_1 u_1 + \cdots + \lambda_k u_k + \mu u = 0$. If $\mu \neq 0$, solving for u gives

$$u = -\frac{1}{\mu}(\lambda_1 u_1 + \cdots + \lambda_k u_k),$$

   so u would be in $\operatorname{span}(u_1, \dots, u_k)$, which it is not. So $\mu = 0$. The relation is then $\lambda_1 u_1 + \cdots + \lambda_k u_k = 0$, and all $\lambda_i = 0$ because $u_1, \dots, u_k$ is independent.
4. So $u_1, \dots, u_k$ is independent and spans U: it is a basis of U. Hence U is finite-dimensional and $\dim U = k \leq \dim V$.
5. *Equality.* If $\dim U = \dim V$, then $u_1, \dots, u_k$ is an independent list of $\dim V$ vectors in V, so it is a basis of V (Prop. 13). Then every vector of V is a combination of $u_1, \dots, u_k$, and those combinations are in U. So $V \subseteq U$, that is, $U = V$. Conversely, if $U = V$ the dimensions are equal.

□

Proposition (LADR 2.33: every subspace has a complement)
If U is a subspace of a finite-dimensional V, there is a subspace W with $V = U \oplus W$.

Proof. 1. U has a basis $u_1, \dots, u_k$ (Prop. 14). It is independent in V, so by Prop. 11 it extends to a basis $u_1, \dots, u_k, w_1, \dots, w_l$ of V. Let $W = \operatorname{span}(w_1, \dots, w_l)$.

2. $U + W = V$**.** Every $v \in V$ is $\sum a_i u_i + \sum b_j w_j$, with $\sum a_i u_i \in U$ and $\sum b_j w_j \in W$.
3. $U \cap W = \{0\}$**.** If $x \in U \cap W$, then $x = \sum a_i u_i$ and $x = \sum b_j w_j$, so $\sum a_i u_i - \sum b_j w_j = 0$. The whole list is a basis of V, hence independent, so all a_i and b_j are 0, and $x = 0$.

□

The complement is not unique: in $\mathbb{R}^2$, $U = \operatorname{span}\big((1,0)\big)$ has the complements $\operatorname{span}\big((0,1)\big)$ and $\operatorname{span}\big((1,1)\big)$.

Example 18: the basis $1, x-a, \dots, (x-a)^n$ of $\mathcal{P}_n(\mathbb{R})$. Independent. Suppose $c_0 + c_1(x-a) + \cdots + c_n(x-a)^n = 0$ with some $c_k \neq 0$, and let k be the largest such index. Expanding, $(x-a)^j = x^j + (\text{lower powers})$, so the coefficient of x^k on the left is $c_k \neq 0$, which contradicts the independence of the powers of x. So all $c_k = 0$.

A basis. There are $n + 1 = \dim \mathcal{P}_n(\mathbb{R})$ of them, so they form a basis (Prop. 13).

The coordinates. Write $p = c_0 + c_1(x-a) + \cdots + c_n(x-a)^n$. Differentiating k times and putting $x = a$ leaves only $k!\, c_k$: the terms with $j < k$ are differentiated to 0, and the terms with $j > k$ keep a factor $x - a$. So $c_k = p^{(k)}(a)/k!$, which is Taylor's formula:

$$p(x) = p(a) + p'(a)(x-a) + \frac{p''(a)}{2}(x-a)^2 + \cdots + \frac{p^{(n)}(a)}{n!}(x-a)^n.$$

Theorem (Prop. 15, 16 / LADR 2.43: dimension of a sum)

For subspaces U, W of a finite-dimensional space:

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W).$$

More precisely, if $v_1, \ldots, v_r$ is a basis of $U \cap W$, it extends to a basis $v_1, \ldots, v_r, u_1, \ldots, u_k$ of U and to a basis $v_1, \ldots, v_r, w_1, \ldots, w_l$ of W, and then $v_1, \ldots, v_r, u_1, \ldots, u_k, w_1, \ldots, w_l$ is a basis of $U+W$.

The idea

For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$: adding the two sizes counts the common part twice. The same happens with bases, provided the bases of U and W are chosen so that their common part is a basis of $U \cap W$. That is why the proof *starts* from a basis of $U \cap W$ and extends it in two directions. The three groups together span $U+W$. For independence, ask whether the u-part and the w-part of a relation could cancel each other: move the w-part to the other side; then one vector lies in both U and W, so it is a combination of the v's alone.

Proof. 1. Extend $v_1, \ldots, v_r$ to bases of U and of W as stated (Prop. 11 in U and in W).

2. The combined list spans $U+W$: every $u+w$ is a combination of it.

3. Independence. Suppose $\sum \lambda_i v_i + \sum \mu_j u_j + \sum \nu_t w_t = 0$. Then

$$x = \sum \lambda_i v_i + \sum \mu_j u_j = -\sum \nu_t w_t.$$

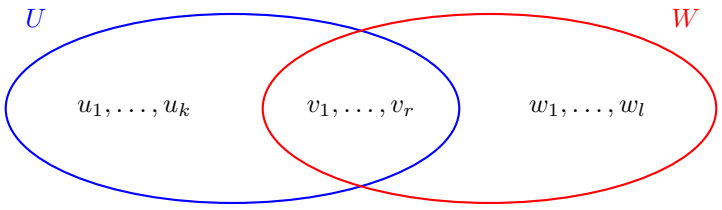
The first expression is in U, the second in W, so $x \in U \cap W$.

4. So $x = \sum \alpha_i v_i$ for some α_i. Then $\sum \alpha_i v_i + \sum \nu_t w_t = 0$ is a relation in the basis of W, so every $\nu_t = 0$.

5. The original relation is now $\sum \lambda_i v_i + \sum \mu_j u_j = 0$, a relation in the basis of U, so every $\lambda_i = 0$ and $\mu_j = 0$.

6. Count: $\dim(U \cap W) = r$, $\dim U = r+k$, $\dim W = r+l$, $\dim(U+W) = r+k+l$.

□



$$\dim(U+W) = r+k+l = (r+k)+(r+l)-r$$

A basis of $U \cap W$ (middle) extended to a basis of U (left two groups) and of W (right two groups). All three groups together are a basis of $U+W$.

Proposition (Complement criterion)

For subspaces U, W of a finite-dimensional V, any two of the following imply the third (and then $V = U \oplus W$):

(a) $\dim U + \dim W = \dim V$ (b) $U \cap W = \{0\}$ (c) $U + W = V$.

Proof. All three cases use the formula $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$. Note that (b) says $\dim(U \cap W) = 0$, and (c) says $\dim(U+W) = \dim V$ (a subspace of V with the same dimension as V is V, Prop. 14).

1. *(a) and (b) give (c).* The formula gives $\dim(U+W) = \dim U + \dim W - 0 = \dim V$. So $U + W = V$.

2. *(a) and (c) give (b).* The formula gives $\dim V = \dim V - \dim(U \cap W)$. So $\dim(U \cap W) = 0$, and a space of dimension 0 is $\{0\}$.

3. *(b) and (c) give (a).* The formula gives $\dim V = \dim U + \dim W - 0$.

When all three hold, $U + W = V$ and $U \cap W = \{0\}$, which is the definition of $V = U \oplus W$ (W2). □

Example 19: two planes in $\mathbb{F}^3$. $U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$, $W = \operatorname{span}\big((1,2,2),(2,2,1)\big)$.

1. Is $(1,2,2) \in U$? $(1,2,2) = \lambda(1,1,0) + \mu(0,1,1)$ needs $\lambda = 1$ (first coordinate) and $\mu = 2$ (third), but then the second coordinate is $3 \neq 2$. No.
2. So $(1,1,0),(0,1,1),(1,2,2)$ are independent (last-vector test): three independent vectors in $\mathbb{F}^3$, a basis (Prop. 13). Hence $U + W = \mathbb{F}^3$.
3. $\dim U = \dim W = 2$: in each pair neither vector is a multiple of the other (compare coordinates). So $\dim(U \cap W) = 2 + 2 - 3 = 1$: the planes meet in a line.
4. To find the line, solve $a(1,1,0) + b(0,1,1) = c(1,2,2) + d(2,2,1)$. Comparing coordinates,

$$a = c + 2d, \qquad a + b = 2c + 2d, \qquad b = 2c + d.$$

Adding the first and third: $a + b = 3c + 3d$, so $c + d = 0$. Take $c = 1$, $d = -1$: the vector is $(1,2,2) - (2,2,1) = (-1,0,1)$. So $U \cap W = \operatorname{span}\big((-1,0,1)\big)$.

The question answered. Yes. $U + W \subseteq \mathbb{R}^3$ has dimension at most 3, so

$$\dim(U \cap W) = \dim U + \dim W - \dim(U + W) \geq 2 + 2 - 3 = 1.$$

Example 19 shows that the dimension is exactly 1 and finds the line.

Example 20: three subspaces. Apply the formula to $U_1 + U_2$ and U_3, then to U_1 and U_2:

$$\dim(U_1 + U_2 + U_3) = \dim U_1 + \dim U_2 + \dim U_3 - \dim(U_1 \cap U_2) - \dim\big((U_1 + U_2) \cap U_3\big).$$

There is no formula using only the pairwise intersections: for three distinct lines in $\mathbb{F}^2$ all pairwise intersections are $\{0\}$, so the pattern of inclusion–exclusion would give $1 + 1 + 1 - 0 - 0 - 0 = 3$, but $\dim(U_1 + U_2 + U_3) = 2$.

Proposition (Prop. 17 / LADR 3.94)
If the sum $U_1 \oplus \cdots \oplus U_m$ is direct, then $\dim(U_1 \oplus \cdots \oplus U_m) = \dim U_1 + \cdots + \dim U_m$.

Proof. Induction on m.

1. **$m = 1$.** The formula reads $\dim U_1 = \dim U_1$.
2. **From $m-1$ to m.** Let $S = U_1 + \cdots + U_{m-1}$. This sum is direct, because its conditions $U_k \cap (U_1 + \cdots + U_{k-1}) = \{0\}$ for $k \leq m - 1$ are among the conditions for $U_1 + \cdots + U_m$ (W2). By the induction hypothesis, $\dim S = \dim U_1 + \cdots + \dim U_{m-1}$.
3. By Prop. 16 applied to S and U_m,

$$\dim(S + U_m) = \dim S + \dim U_m - \dim(S \cap U_m),$$

and $S \cap U_m = \{0\}$ is the last condition in the definition of a direct sum (W2). Adding gives $\dim(U_1 + \cdots + U_m) = \dim U_1 + \cdots + \dim U_m$.

□

Computing with lists in $\mathbb{F}^n$.

1. *Independent?* Set a general combination equal to 0 and solve the resulting linear equations for the coefficients; the list is independent exactly when the only solution is 0 (as in R1). In $\mathbb{F}^n$ the bookkeeping is row reduction from lower-division linear algebra (cited): with the vectors as columns, the list is independent exactly when every column has a pivot. The pivot columns of the original matrix form a basis of the span, because row operations do not change which combinations of the columns are 0, and in the reduced matrix each non-pivot column is a combination of the pivot columns before it, while the pivot columns are different standard basis vectors.
2. *Basis of $U + W$:* put the two bases together and delete redundant vectors (Prop. 9).

3. *Basis of $U \cap W$:* set a general combination of the basis of U equal to a general combination of the basis of W and solve, as in Example 19. First get its dimension from the formula so you know how many vectors to expect.

Shortcuts

- **Right number of vectors**: with exactly $\dim V$ vectors, check only independence or only spanning. Reason: Prop. 13. Condition: V finite-dimensional and the list has length $\dim V$. Instance: $(1,1), (1,-1)$ are not multiples of each other, so they are independent, so they form a basis of $\mathbb{R}^2$.
- **Too many or too few**: more than $\dim V$ vectors are dependent; fewer than $\dim V$ do not span. Reason: Prop. 10, with a basis as the spanning list, resp. as the independent list. Instance: any 4 polynomials in $\mathcal{P}_2(\mathbb{R})$ are dependent, since $\dim \mathcal{P}_2(\mathbb{R}) = 3$.
- **Intersections forced by dimension**: $\dim(U \cap W) \geq \dim U + \dim W - \dim V$. Reason: Prop. 16 and $\dim(U + W) \leq \dim V$ (Prop. 14). Instance: two 3-dimensional subspaces of $\mathbb{R}^5$ share a nonzero vector, since $3 + 3 - 5 = 1$.
- **Direct sum by counting**: if $U + W = V$ and $\dim U + \dim W = \dim V$, the sum is direct, with no need to compute $U \cap W$. Reason: complement criterion. Instance: HW2 P5.
- **Equality of subspaces**: $U' \subseteq U$ and $\dim U' = \dim U$ give $U' = U$. Reason: Prop. 14 applied inside U. Instance: $U' = \operatorname{span}\big((1,2,1),(1,0,-1)\big)$ lies in $U = \operatorname{span}\big((1,1,0),(0,1,1)\big)$, because $(1,2,1) = (1,1,0) + (0,1,1)$ and $(1,0,-1) = (1,1,0) - (0,1,1)$; both have dimension 2, so $U' = U$.
- **Basis of a subspace given by equations**: solve for some coordinates in terms of the others (the free parameters); setting one free parameter to 1 and the rest to 0 gives one basis vector per free parameter. Reason: every solution is the combination of these vectors with its own parameter values (spanning), and the j-th vector is the only one with a nonzero entry in the j-th free coordinate (independence). Instance: $x_1 + x_2 + x_3 = 0$ in $\mathbb{R}^3$ gives $x_1 = -x_2 - x_3$ with x_2, x_3 free, the basis $(-1,1,0), (-1,0,1)$, and dimension 2. The same method works for conditions on polynomial coefficients (R3).
- **Extending to a basis in $\mathbb{F}^n$**: append $e_1, \ldots, e_n$ and delete each vector that lies in the span of the vectors before it. Reason: Prop. 11 with the spanning list $e_1, \ldots, e_n$. Instance: for $(1,1,0)$: keep e_1 (not a multiple of $(1,1,0)$), delete $e_2 = (1,1,0) - e_1$, keep e_3 (third coordinate 1). Basis: $(1,1,0), e_1, e_3$.

Traps

- $\dim(U + W) = \dim U + \dim W$ only when $U \cap W = \{0\}$: for $U = W$ a line in $\mathbb{R}^2$, $\dim(U + W) = 1 \neq 2$.
- $\dim(U_1 + U_2 + U_3)$ has no inclusion–exclusion formula with pairwise intersections: three distinct lines in $\mathbb{R}^2$ give $1 + 1 + 1 - 0 - 0 - 0 = 3$, but $\dim(U_1 + U_2 + U_3) = 2$.
- A basis of V need not contain a basis of a given subspace U: e_1, e_2 contains no basis of $\operatorname{span}\big((1,1)\big)$.
- The dimension depends on the field: $\dim_{\mathbb{C}} \mathbb{C} = 1$, but $\dim_{\mathbb{R}} \mathbb{C} = 2$ (basis $1, i$).
- "$U \subseteq V$ with bases of the same size $\Rightarrow U = V$" and "an independent list as long as a basis is a basis" need finite dimension. In $\mathcal{P}(\mathbb{R})$ the subspace $U = \{p : p(0) = 0\}$ has the basis $x, x^2, x^3, \ldots$, which matches the basis $1, x, x^2, \ldots$ of $\mathcal{P}(\mathbb{R})$ term by term ($x^k \leftrightarrow x^{k+1}$). Yet $1 \notin U$, so $x, x^2, \ldots$ is an infinite independent set that is not a basis of $\mathcal{P}(\mathbb{R})$, and $U \neq \mathcal{P}(\mathbb{R})$.

§3 Problem set

HW2 P1 (LADR 2A.11). Prove or give a counterexample: if $v_1, \ldots, v_n$ and $w_1, \ldots, w_n$ are linearly independent lists in V, then $v_1 + w_1, \ldots, v_n + w_n$ is linearly independent.

Hint: try $n = 1$.

HW2 P2. Let $\mathbb{F}^\infty$ be the space of all sequences $(a_1, a_2, \dots)$, $a_j \in \mathbb{F}$, and e_j the sequence with 1 in position j and 0 elsewhere. (a) Verify that $\mathbb{F}^\infty$ is a vector space. (b) Prove that it is infinite-dimensional. (c) Prove that $e_1, e_2, \dots$ do not span $\mathbb{F}^\infty$.

Hint: (b) Prop. 10: a finite spanning list bounds the length of independent lists. (c) a finite combination has only finitely many nonzero entries.

HW2 P3 (LADR 2B.5). Suppose V is finite-dimensional and U, W are subspaces with $V = U + W$. Prove that V has a basis all of whose vectors belong to $U \cup W$.

Hint: a spanning list contains a basis (Prop. 9).

HW2 P4. (a) Prove that every nonzero proper subspace of $\mathbb{F}^2$ has the form $\operatorname{span}(v)$ for some $v \neq 0$. (b) Let $v, w \in \mathbb{F}^2$. Prove that v, w is a basis of $\mathbb{F}^2$ if and only if neither is a scalar multiple of the other.

Hint: dimensions of subspaces (Prop. 14) and Prop. 13.

HW2 P5 (LADR 2C.12). Suppose U, W are subspaces of $\mathbb{R}^8$ with $\dim U = 3$, $\dim W = 5$ and $U + W = \mathbb{R}^8$. Prove that $\mathbb{R}^8 = U \oplus W$.

Hint: Prop. 16.

HW2 P6. Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}$. (a) Prove that U is a subspace. (b) Find a basis of U and determine $\dim U$ directly, without rank–nullity.

Hint: write p in the basis $1, (x-1), \dots, (x-1)^4$ of Example 18.

R1 (Practice Midterm 1, Question 1). Let $u_1 = (1, 1, 0, 0)$, $u_2 = (0, 1, 1, 0)$, $w_1 = (0, 0, 1, 1)$, $w_2 = (1, 0, 0, -1)$, $U = \operatorname{span}(u_1, u_2)$, $W = \operatorname{span}(w_1, w_2)$. (a) Prove that $\mathbb{R}^4 = U \oplus W$. (b) Write $(2, 1, 3, 0)$ as $u + w$ with $u \in U$, $w \in W$.

Hint: four vectors in $\mathbb{R}^4$: show u_1, u_2, w_1, w_2 independent (Prop. 13), then use the complement criterion; for (b) solve for the four coefficients.

R2 (Midterm 1 review, Exercise 1). Let U be a subspace of a finite-dimensional V. Prove that there is a subspace W with $U \oplus W = V$.

Hint: extend a basis of U to a basis of V (Prop. 11) and take the span of the added vectors.

R3 (Midterm 1 review, Exercise 3). Let $V = \{p \in \mathcal{P}_2(\mathbb{R}) : p'(1) = p(2)\}$. Construct a basis of V and prove it is a basis.

Hint: write $p = a + bx + cx^2$, turn $p'(1) = p(2)$ into one linear equation in a, b, c, and solve with free parameters.

R4 (Midterm 1 review, Exercise 15). Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, let $v_1, \dots, v_n \in V$, and for each i let $w_i = \sum_{j=1}^{i} j v_j$. Prove that $\operatorname{span}(v_1, \dots, v_n) = \operatorname{span}(w_1, \dots, w_n)$.

Hint: equal spans (Shortcuts of Section 2.1): each w_i is a combination of the v's; for the other direction use $w_i - w_{i-1} = i v_i$.